

INDEPENDENT MARKET & BENCHMARK NOTE

Project Helios Solar & Storage - Illustrative POC Support Document

IMPORTANT: This document and all figures are entirely synthetic and were created solely for a proof-of-concept demonstration.

1. Scope and limitations

This synthetic note provides a compact market and benchmark reference for the POC. It is designed to test document retrieval, metric extraction, benchmark comparison and identification of unsupported assumptions. It is not investment advice and does not represent an actual market forecast.

2. Illustrative market outlook

Year	GDP growth	Inflation	THB/USD	Merchant power price
2026	3.0%	1.5%	35.8	\$61/MWh
2027	3.2%	1.8%	36.1	\$63/MWh
2028	3.3%	2.0%	36.5	\$64/MWh
2029	3.4%	2.1%	36.8	\$65/MWh
2030	3.2%	2.0%	37.0	\$66/MWh

The merchant-price path above is synthetic. The project submission references a similar curve but does not append an independent source or methodology.

3. Solar and storage benchmark ranges

Metric	Illustrative peer range	Helios submission	Observation
Solar EPC cost	USD 0.55-0.75m/MW	USD 0.61m/MW excluding BESS	Within range
BESS equipment	USD 0.20-0.25m/MWh	USD 0.26m/MWh	Upper end / above range
Grid connection	5%-8% of total cost	8.7%	Above range
Contingency	3%-5% of construction cost	6%+ of total project cost	High
Corporate overhead	1%-3% of total cost	4.2%	High
Net capacity factor	22%-25%	24.7%-25.5% across documents	Plausible but inconsistent

4. Key benchmark challenges

- Reconcile the project cost across the deck, memorandum and workbook.
- Support the BESS equipment cost and augmentation assumptions with firm supplier quotations.
- Validate grid reinforcement scope and whether any utility contribution is available.
- Demonstrate that advisory and corporate-overhead scopes do not overlap.
- Provide an independent merchant curve and downside confidence range.

SYNTHETIC POC DATA - NOT FOR INVESTMENT DECISION

- Quantify the return impact of delayed COD, combined downside and battery augmentation.

5. Public comparator references available in the POC repository

- ADB Gulf Solar and Solar with BESS project documents.
- ADB utility-scale BESS and floating-solar project pages.
- IFC solar, renewables-platform and BESS disclosures.
- World Bank Private Participation in Infrastructure project database.
- NREL Annual Technology Baseline model parameters.
- SEC annual reports and XBRL facts for selected renewable-energy companies.